

PRST (16/09/2021).

Mgf of $X_1 \rightarrow M_1(t)$
Mgf of $X_2 \rightarrow M_2(t)$.

Mgf of $X_1 + X_2$ if X_1 and X_2 are independent rvs
 $E(e^{t(X_1+X_2)}) = E(e^{tX_1} e^{tX_2})$
 $= E(e^{tX_1}) E(e^{tX_2})$

$$F(x, y) = P(X \leq x, Y \leq y)$$

$$F(-\infty, y) = 0$$

$$F(x, -\infty) = 0$$

$$F(x, +\infty) = G_1(x)$$

$$F(+\infty, y) = H(y)$$

$$F(\infty, \infty) = 1$$

$\rightarrow$ marginal distribution functions.

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{4} & 0 \leq x < 2 \\ \frac{1}{2} & 2 \leq x < 4 \\ 1 & x \geq 4 \end{cases}$$

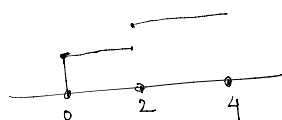
$$P(X=x) = ?$$

$$P(X=0) = \frac{1}{4}$$

$$P(X=2) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$P(X=4) = 1 - \frac{1}{2} = \frac{1}{2}$$

$$F(2) = P(X=0) + P(X=2)$$



Ex. X, Y rvs discrete
 $P(X=2) = \frac{1}{3}$
 $P(Y=3) = \frac{1}{3}$
 $P(Y=4) = \frac{1}{3}$
 $P(X=1, Y=2) = \frac{1}{4}$
 $P(X=2, Y=3) = \frac{1}{12}$

① find the conditional mean of Y given $X=2$.
 $P(Y=2 | X=2) = \frac{P(Y=2, X=2)}{P(X=2)}$
 $P(Y=3 | X=2)$
 $P(Y=4 | X=2)$

X \ Y	2	3	4
1	$\frac{1}{4}$		
2		$\frac{1}{12}$	$\frac{1}{3}$

$$E(Y | X=2) = 2 \cdot P(Y=2 | X=2) + 3 \cdot P(Y=3 | X=2) + 4 \cdot P(Y=4 | X=2)$$

Correlation coefficient between X and Y .

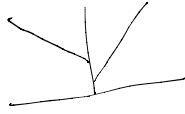
$$r_{XY} = \text{Cor}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}$$

$$\begin{cases} X, Y \text{ cont rvs} \\ E(Y | X=x_0) = \int_{-\infty}^{\infty} y f(y | x_0) dy \end{cases}$$

$$r_{xy} = \frac{\text{Cor}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{\omega}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

$$-1 \leq r \leq 1.$$

$$X = a + bY$$



Eg. X, Y cont rvs
joint pdf $f(x, y) = \begin{cases} 2e^{-(x+y)} & 0 < x < y < \infty \\ 0 & \text{o.w.} \end{cases}$

and conditional distributions of X and Y .
Obtain marginal

sol: $g(x) = \int_x^\infty 2e^{-(x+y)} dy = 2e^{-x}, x \geq 0$
pdf of X
pdf of Y $h(y) = \int_0^y 2e^{-(x+y)} dx = 2e^{-y}(1 - e^{-y}), y \geq 0$

$$f(x/y) = \frac{f(x, y)}{h(y)}$$

$$f(y/x) = \frac{f(x, y)}{g(x)}$$